

Original Research

FuseLinker: Leveraging LLM's pre-trained text embeddings and domain knowledge to enhance GNN-based link prediction on biomedical knowledge graphs

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ARTICLE INFO

Keywords:

Link Prediction
Knowledge graph
Graph Neural Network
Large Language Model
Drug Repurposing

ABSTRACT

Objective: To develop the FuseLinker, a novel link prediction framework for biomedical knowledge graphs (BKGs), which fully exploits the graph's structural, textual and domain knowledge information. We evaluated the utility of FuseLinker in the graph-based drug repurposing task through detailed case studies.

Methods: FuseLinker leverages fused pre-trained text embedding and domain knowledge embedding to enhance the graph neural network (GNN)-based link prediction model tailored for BKGs. This framework includes three parts: a) obtain text embeddings for BKGs using embedding-visible large language models (LLMs), b) learn the representations of medical ontology as domain knowledge information by employing the Poincaré graph embedding method, and c) fuse these embeddings and further learn the graph structure representations of BKGs by applying a GNN-based link prediction model. We evaluated FuseLinker against traditional knowledge graph embedding models and a conventional GNN-based link prediction model across four public BKG datasets. Additionally, we examined the impact of using different embedding-visible LLMs on FuseLinker's performance. Finally, we investigated FuseLinker's ability to generate medical hypotheses through two drug repurposing case studies for Sorafenib and Parkinson's disease.

Results: By comparing FuseLinker with baseline models on four BKGs, our method demonstrates superior performance. The Mean Reciprocal Rank (MRR) and Area Under receiver operating characteristic Curve (AUROC) for KEGG50k, Hetionet, SuppKG and ADInt are 0.969 and 0.987, 0.548 and 0.903, 0.739 and 0.928, and 0.831 and 0.890, respectively.

Conclusion: Our study demonstrates that FuseLinker is an effective novel link prediction framework that integrates multiple graph information and shows significant potential for practical applications in biomedical and clinical tasks. Source code and data are available at <https://github.com/YKXia0/FuseLinker>.

1. Introduction

A knowledge graph (KG) is a graph structure designed to accumulate and convey knowledge about the real world, with nodes representing entities of interest and edges representing potential different relationships between these entities [1]. Biomedical Knowledge Graph (BKG) is a structured representation of scientific knowledge in the field of biomedicine, depicting the relations among various biological, clinical, and medical entities [2]. For example, a triple might convey information about how a certain gene influences a particular disease or how a

specific drug interacts with a certain gene. However, BKGs are often incomplete, primarily due to the absence of edges that should describe factual relationships. This phenomenon can be attributed to two main factors [3]: the incompleteness of information in the data sources used to construct the BKG, and the presence of relationships between two entities that remain undiscovered or unknown to humanity. Link prediction on KG, also known as knowledge graph completion (KGC), aims to predict such missing or novel relationships using the existing information on the KG [4]. In the context of BKGs, link prediction is instrumental in uncovering previously unknown associations or interactions among

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<https://doi.org/10.1016/j.jbi.2024.104730>

Received 16 July 2024; Received in revised form 20 September 2024; Accepted 23 September 2024

Available online 24 September 2024

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drugs, genes, disease entities, etc. It serves as a crucial method for generating new hypotheses and can be applied in various downstream biomedical tasks [5], such as drug repurposing [6,7].

Link prediction is a commonly studied problem in data mining and machine learning on graphs, and there are numerous algorithms and models that can be employed to tackle it. Initially, simpler heuristics were popular for assessing the similarity between pairs of nodes based on local topological features, including measuring the number of common neighbors and utilizing the Jaccard coefficient [8]. With the advancement of representation learning, graph embedding has increasingly become a predominant approach for link prediction. Random walk-based methods such as DeepWalk [9] and Node2Vec [10] have been developed to encapsulate the global topology of graphs within node embeddings. Conversely, link prediction methods based on translation models, such as TransE [11], and semantic matching models, including DistMult [12] and ComplEx [13], are designed to explicitly model the triples within graphs, focusing on deriving embeddings that capture the intricate relationships and entities within the KG. The emergence of graph neural networks (GNNs) [14] has revolutionized link prediction. GNN Models such as graph convolutional networks (GCNs) [15] and graph attention networks (GATs) [16] effectively utilize both node feature information and graph structures, offering a robust framework for discovering complex patterns in graph data. Recently, leveraging pre-trained language models for link prediction has emerged as a novel approach, such as KG-BERT [17] and SimKGC [18]. These methods convert triples in KG into natural text, feed them into pre-trained language models to obtain triple scores or directly generate results, and then optimize the model parameters by fine-tuning. Compared to traditional graph embedding methods, these approaches fully utilize the textual information of KGs, such as the names, description and textual attributes of nodes and relations, but they may lose graph structure information to a certain extent.

Fully leveraging both graph structure and text information has the potential to significantly enhance the performance of link prediction models. Recently, the practice of combining large language models (LLMs) with GNNs has emerged as a promising approach to achieve this integration [19]. There are two main paradigms: LLM-as-enhancer and LLM-as-predictor. LLM-as-enhancer approaches involve improving the quality of textual attributes or node embeddings with the help of powerful LLMs. Meanwhile, LLM-as-predictor approaches leverage LLMs to make predictions directly across a wide range of graph-related tasks, including node classification and link prediction [20,21]. Lately, there have been significant advances in LLMs. After extensive pre-training on large corpora, LLMs can achieve outstanding performance in downstream tasks such as machine translation, text classification, and artificial intelligence chatbot, utilizing methods like fine-tuning, few-shot learning, and instruction tuning [22,23]. LLMs can be categorized based on their architectural structure into three types: (1) encoder-only LLMs, (2) encoder-decoder LLMs, and (3) decoder-only LLMs [21]. Encoder-only models, such as BERT and RoBERTa, focus solely on encoding sentences to discern patterns between words. Encoder-decoder models, like T5 and GLM, feature a decoder module that generates output text from encoded inputs. Many state-of-the-art LLMs, including LLAMA and GPT-4, are decoder-only, relying exclusively on their decoder modules for text generation.

Different from general domain KGs, BKGs and link prediction on BKGs have their own characteristics. On the one hand, BKGs often include entities, such as genes and proteins, without sufficient descriptions and text attributes. On the other hand, BKGs are rich in biomedical domain knowledge, such as molecular structure and medical ontology, which provides more natural biology medicine and clinical information. Additionally, for link prediction on BKGs, especially in practical applications like drug repurposing, the outcomes for each corrupted triple are not limited to a single entity. That is, in real-world scenarios, a drug may treat multiple diseases, and a disease might be treatable by various drugs. Therefore, it is essential that developing link

prediction methods for BKGs should be capable of generating predictions for multiple candidates. In this study, we propose a novel approach, FuseLinker, that enhances the GNN-based link prediction model on BKG by integrating fused pre-trained text information and domain knowledge. FuseLinker consists of three key modules: (1) Text embedding acquisition for graph nodes using embedding-visible LLMs; (2) Customization of domain knowledge embedding from a hierarchical medical ontology, UMLS Semantic Network [24], which includes learning the ontology representations by a multi-relational Poincaré model and mapping BKG nodes to the ontology; (3) A multi-relation GNN model with a fusion strategy, fusing pre-trained text embeddings and domain knowledge embeddings for initialization and then undergoes training on the link prediction task to learn graph structural information. We compared our method, FuseLinker, with traditional embedding-based link prediction methods—TransE, DistMult, ComplEx, and vanilla RGCN—across four representative BKG datasets: KEGG50k [25], Hetionet [26], SuppKG [27], and ADInt [7]. We also evaluated the performance of FuseLinker using different types of embedding-visible LLMs. To demonstrate our method's practical application, we conducted two drug repurposing case studies and verified the predicted candidates against clinical ground truths and published literature. Our findings indicate that FuseLinker delivers state-of-the-art (SOTA) performance on BKGs and demonstrates practical effectiveness in biomedical and clinical fields, which suggests FuseLinker's potential to solve other real complex biomedical and clinical tasks, such as detecting adverse drug reactions and drug-drug interactions.

Statement of significance.

Problem or Issue:

Traditional Link prediction models often underperform on biomedical knowledge graphs (BKGs).

What is Already Known:

Recently, approaches integrating graph textual information and graph structural information have been increasingly utilized for general graph learning tasks in the general domain, leveraging the textual comprehension power of LLMs alongside the structural understanding provided by graph learning techniques.

What this Paper Adds:

This paper introduces FuseLinker, a novel link prediction framework designed for BKGs that effectively harnesses and integrates graphs' structural, textual, and domain knowledge information. Notably, FuseLinker utilizes medical ontology as domain knowledge to enrich the information representations of the graph, which is tailor-made for the unique characteristics of BKGs.

2. Methods

FuseLinker focuses on utilizing the text information embedding and domain knowledge embedding to enhance the graph representation learning of GNN, thereby improving performance on the link prediction tasks. In contrast to conventional methods that depend solely on graph structural features, FuseLinker incorporates rich textual information and domain-specific knowledge. This integration enables a deeper semantic understanding of node relationships, leading to a more robust and context-aware prediction model. Fig. 1 shows the framework of FuseLinker, which comprises three main modules: (i) generating graph's text embeddings by LLMs, (ii) domain knowledge information mapping and hierarchical embedding learning, and (iii) fusing embeddings and training the GNN-based model for link prediction tasks. These modules collectively enhance the model's ability to discern and predict complex relationships within the graph. Before detailing each module in this section, we provide study settings and descriptions of datasets used.

2.1. Study settings

A KG can be represented as a labeled directed multigraph $KG = (E, R, G)$. For BKG, E is the set of nodes representing biomedical entities, R is

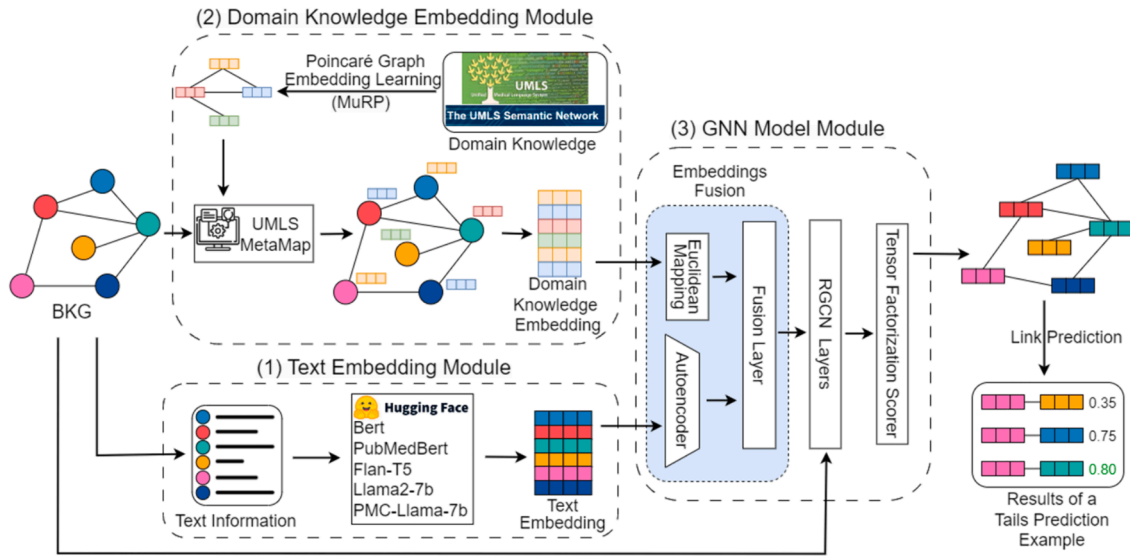


Fig. 1. The framework of FuseLinker with three modules: (1) Text Embedding Module, (2) Domain Knowledge Embedding Module, and (3) GNN Model Module.

the set of edges describing all possible relationships between these biomedical entities, and $G \subseteq E \times R \times E$ is a set of triples (h, r, t) , where h and r are the head and tail entities (nodes), and r is the relation (link) between them.

Link prediction tasks on BKGs aim to infer novel or missing triples through the existing ones. These triples (h', r', t') where $h', t' \in E$ and $r' \in R$ may also describe some facts, but for some reason they don't exist in our knowledge graph, i.e., $(h', r', t') \notin G$. Link prediction can be equivalent to guessing the correct entity or relation to complete $(h, r, ?)$ (tail prediction), $(h, ?, t)$ (edge prediction) or $(?, r, t)$ (head prediction). Further, link prediction for BKGs can be modeled as a machine learning task, which is typically approached through a ranking framework. This framework learns and optimizes a scoring function that effectively ranks true triples higher than false ones by assigning them differential scores based on their likelihood of being correct.

2.2. Datasets

In this study, we adopted four BKGs as our datasets: two general BKG including KEGG50k [25] and Hetionet [26], and two domain specific BKGs, including SuppKG [27] and ADInt [7]. Their detailed statistics were shown in Table S1 in Supplementary Material. KEGG50k is a generated benchmarking BKG based on the kyoto encyclopedia of genes and genomes (KEGG) database [28], primarily used for predicting drug target proteins as a link prediction task on BKG. Hetionet [26] is an integrative knowledge graph that maps biological relationships among genes, diseases, drugs, and etc. It supports drug repurposing and disease gene discovery by providing a network-based structure that enables the exploration of complex biological data for translational research. SuppKG is a BKG in the field of dietary supplements (DS), which contains information and triples about DS contained in the integrated Dietary Supplement Knowledge base (iDISK) [29]. ADInt [7] is a literature-based BKG focused on Alzheimer's disease and its associated biomedical information. It was initially developed and utilized to repurpose pharmaceutical and non-pharmacological interventions for Alzheimer's disease through various link prediction methods. Compared with conventional BKG, ADInt not only includes common entity categories such as drugs, genes, proteins, and diseases but also contains annotated entities for DS and complementary and integrative health (CIH) interventions.

We utilized all four BKGs—KEGG50k, Hetionet, SuppKG, and ADInt—to evaluate and compare the performance of FuseLinker and other baseline models in link prediction tasks. Given that Hetionet was

specifically constructed for drug repurposing purposes, we selected this BKG for our drug repurposing case studies. We retained only the gene, compounds, and disease entities along with their corresponding relationships [30]. The filtered Hetionet comprised 19,930 nodes of 3 types and 562,106 relationships of 13 types. We created inverse relations for the relationships that should be undirected edges in the graph semantically, such as 'associates', 'covaries' and 'interacts'.

2.3. Text information embedding

To obtain the initial representation of the graph nodes, we considered deriving text embeddings of the nodes using LLMs. In the knowledge graph, the text information of a node can originate from the node's name, description or any other textual attributes. Considering the ease of acquisition and standardization of different types of texts, as well as the robust capabilities of LLMs trained on extensive corpora, we opted to use the name of each node as the textual information. This approach allows us to consistently compare the performance of our method using different LLMs across different BKGs.

Given the requirement for easily obtainable text embeddings, we opted not to use embedding-invisible LLMs like GPT-3.5 and GPT-4. Considering that the nodes in BKGs cover a broad spectrum of biological and clinical concepts, including genes and drugs, we selected and added models pre-trained on the biomedical corpus for comparison. Specifically, we chose two encoder-only LLMs (BERT [31] and PubMedBERT [32]), one encoder-decoder LLMs (Flan-T5 [33]), and two decoder-only LLMs (Llama2 [34] and PMC-LLaMA [35]) to assess the impact of embeddings obtained through different LLMs on link prediction performance. Among them, PubMedBERT and PMC-LLaMA are specialized in the biomedical field. Due to different model architectures and design concepts, there are subtle differences in how encoder-only LLMs, encoder-decoder LLMs, and decoder-only LLMs obtain text embedding. For BERT and PubMedBERT, we chose the first token embedding of the last layer output of the model (i.e., the special [CLS] token) to represent the embedding of the entire input sequence. For Flan-T5, we computed the mean of the embeddings from all tokens in the hidden state output of the model's final layer to obtain a representation of the entire input sequence. For Llama2 and PMC-LLaMA, we selected the embedding of the last non-padding token in the input sequence from the final layer's output as the text embedding.

2.4. Domain knowledge embedding

To further enrich the initial representation of graph nodes, we vectorize the domain knowledge in BKG to generate specialized embeddings for each node in a biomedical context. One of BKG's domain knowledge is medical ontology, which is usually a graph composed of a series of validated and persuasive biomedical *meta*-paths. One part of the Unified Medical Language System (UMLS) is the Semantics Network [24] which is a typical medical taxonomy, [36]. The Semantic Network contains 127 semantic types and 54 predicates, describing more than 6,000 medical semantic relations. The Semantic Network can be modeled as a graph, where all nodes are the 127 semantic types, edge types are the 54 predicates, and all triples are the more than 6,000 semantic relations. Subsequently, we can learn its representation. However, what is different from traditional graphs is that UMLS is a hierarchical graph that has hierarchical predicates (such as 'isa' and 'inverse_isa') to represent the parent-child relationship between nodes. In our study, we implemented a Poincaré graph embedding method, multi-relational Poincaré model (MuRP) [37], to learn the graph representation of UMLS Semantic Network.

MuRP is capable of embedding hierarchical multi-relational graphs in the Poincaré ball model of hyperbolic space. Comparing with traditional embedding methods such as TransE and DistMult usually mapping entities and relationships to Euclidean space, the nature of hyperbolic space determines that Poincaré graph embedding is more suitable for representation learning of hierarchical graphs [38–40].

The Poincaré ball $(\mathbb{B}_c^d, g^{\otimes})$ of radius $1/\sqrt{c}$ is a d -dimensional manifold where $\mathbb{B}_c^d = \{\mathbf{x} \in \mathbb{R}^d, c\|\mathbf{x}\|^2 < 1\}$ and $g^{\otimes}$ is a Riemannian metric. The distance between two points $\mathbf{x}, \mathbf{y} \in \mathbb{B}_c^d$, which is measured along a geodesic, can be written as:

$$d_{\mathbb{B}}(\mathbf{x}, \mathbf{y}) = \frac{2}{\sqrt{c}} \tanh^{-1}(\sqrt{c} \|\mathbf{x} \oplus_c \mathbf{y}\|)$$

where $\oplus_c$ is Möbius addition [41]:

$$\mathbf{x} \oplus_c \mathbf{y} = \frac{(1 + 2c\langle \mathbf{x}, \mathbf{y} \rangle + c\|\mathbf{y}\|^2)\mathbf{x} + (1 - c\|\mathbf{x}\|^2)\mathbf{y}}{1 + 2c\langle \mathbf{x}, \mathbf{y} \rangle + c^2\|\mathbf{x}\|^2\|\mathbf{y}\|^2}$$

where $\langle \mathbf{x}, \mathbf{y} \rangle$ is the Euclidean inner product of $\mathbf{x}$ and $\mathbf{y}$. To represent projecting points on the sphere to tangent space, performing matrix transformation in the Euclidean tangent space and projecting back to $\mathbb{B}_c^d$, Möbius matrix-vector multiplication [42] is introduced:

$$\mathbf{M} \otimes_c \mathbf{x} = \exp_0^c(\mathbf{M} \log_0^c(\mathbf{x}))$$

where $\mathbf{M} \in \mathbb{R}^{d \times k}$, $\log_0^c(\cdot)$ is logarithmic map at $\mathbf{0} \in \mathbb{B}_c^d$ and $\exp_0^c(\cdot)$ is exponential map at $\mathbf{0} \in \mathbb{B}_c^d$.

Thus, The score function of MuRP can be defined as [37]:

$$\begin{aligned} s(\mathbf{h}, \mathbf{r}, \mathbf{t}) &= -d_{\mathbb{B}}(\mathbf{h}^{(r)}, \mathbf{t}^{(r)})^2 + b_h + b_t \\ &= -d_{\mathbb{B}}(\exp_0^c(\mathbf{R} \log_0^c(\mathbf{h})), \mathbf{t} \oplus_c \mathbf{r})^2 + b_h + b_t \end{aligned}$$

where $\mathbf{h}, \mathbf{t} \in \mathbb{B}_c^d$ are hyperbolic embeddings of head nodes and tail nodes and $\mathbf{r} \in \mathbb{B}_c^d$ is a hyperbolic translation vector of their relation. $\mathbf{h}^{(r)} \in \mathbb{B}_c^d$ and $\mathbf{t}^{(r)} \in \mathbb{B}_c^d$ represent the $\mathbf{h}$ and $\mathbf{t}$ after applying the respective relation-specific transformations, which are customized for multi-relational graphs. The relation-adjusted head node embedding $\mathbf{h}^{(r)} \in \mathbb{B}_c^d$ is obtained by Möbius matrix-vector multiplication with by the diagonal relation matrix $\mathbf{R} \in \mathbb{R}_c^{d \times d}$, and the relation-adjusted tail node embedding $\mathbf{t}^{(r)} \in \mathbb{B}_c^d$ is obtained by Möbius addition of the relation $\mathbf{r}$ to the $\mathbf{t}$.

After learning the hierarchical graph representation of UMLS Semantic Network, we map the nodes in BKG to the semantic types in UMLS Semantic Network through MetaMap [43], a Natural language

processing (NLP) tool to map biomedical text to the UMLS Meta-thesaurus [44] or to identify Metathesaurus concepts referred to in nature text equivalently, to obtain the domain knowledge embeddings of each node in BKG. Following the mapping of nodes from KEGG50k, Hetionet, SuppKG, and ADInt, the proportions successfully mapped to UMLS semantic types were 94.6 %, 100.0 %, 100.0 %, and 100.0 %, respectively. For nodes in a BKG that could not be mapped (here only KEGG50k), we assigned the average domain knowledge embeddings of successfully mapped nodes within the respective graph to them.

2.5. GNN model structure

In our GNN model module, we first fuse the obtained text embeddings and the domain knowledge embeddings. Then, the fused embedding is passed to the RGCN layers and the GNN-based model is trained for the link prediction task.

2.5.1. Embeddings fusion

Fusing text embeddings with domain knowledge embeddings first requires addressing two primary challenges: (1) For text embeddings obtained from LLMs, the embedding dimension is not adjustable and depends on the token embedding size of the selected LLMs (for instance, embedding from BERT is 768 and from Llama2 is 4096). (2) The domain knowledge embedding obtained by Poincaré graph embedding is in hyperbolic space, while the BKG representations learned by text embedding and GCN are both in Euclidean space.

For problem (1), we employed an autoencoder to resize the text embeddings so that their dimensions align with that of the embedding in the GNN input layer:

$$\mathbf{E}(\mathbf{x}) = \text{BN}_2(\text{Linear}_2(\text{Dropout}(\text{ReLU}(\text{BN}_1(\text{Linear}_1(\mathbf{x}))))$$

where $\text{Linear}_1(\cdot)$ and $\text{Linear}_2(\cdot)$ denote the linear transformations, and $\text{BN}_1(\cdot)$ and $\text{BN}_2(\cdot)$ denote the batch normalization layers. We chose ReLU as the activation function and added a Dropout layer to reduce overfitting. In this study, we only utilized the encoder module.

For problem (2), we added a learnable fully connected layer to map the Poincaré embedding from hyperbolic space to Euclidean space and make the dimensions consistent with the RGCN input layer:

$$\mathbf{x}_{\text{Euclidean}} = \mathbf{W} \mathbf{x}_{\text{Hyperbolic}} + \mathbf{b}$$

where $\mathbf{x}_{\text{Hyperbolic}}$ represents the Poincaré embeddings, $\mathbf{x}_{\text{Euclidean}}$ denotes the transformed embeddings in Euclidean space, $\mathbf{W}$ is the weight matrix of the transformation layer, and $\mathbf{b}$ is the bias vector.

The parameters of the autoencoder and fully connected layer can be optimized during GNN model training. Since the updated text embedding and domain knowledge embedding have the same dimension size, we were able to fuse the two embedding matrices using their weighted average:

$$\text{embedding}_{\text{fusion}} = w^* \text{embedding}_{\text{text}} + (1 - w)^* \text{embedding}_{\text{knowledge}}$$

w is a hyperparameter of FuseLinker.

2.5.2. GNN encoder and link prediction decoder

After obtaining the text embedding and domain knowledge embedding of each biomedical entity, we further leveraged GNNs to learn the graph's representation based on its structure information.

To be used for heterogeneous graphs such as BKG, we selected the Relational Graph Convolutional Network (RGCN) [45] as our GNN layers. The information propagation and Neighborhood Aggregation model for RGCN calculating the forward-pass update of nodes can be defined as [45]:

$$\mathbf{x}_i^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{I}_i^r} \frac{1}{c_{i,r}} \mathbf{W}_r^{(l)} \mathbf{x}_j^{(l)} + \mathbf{W}_0^{(l)} \mathbf{x}_i^{(l)} \right)$$

where $\mathbf{x}_i^{(l)} \in \mathbb{R}^{d^{(l)}}$ is the hidden state of i -th nodes in the l -th layer of the neural network; $\mathcal{R}$ is the relations set and $\mathcal{N}_i^r$ denotes the neighbor set of i -th node under relation $r \in \mathcal{R}$; $\mathbf{W}_r^{(l)}$ and $\mathbf{W}_0^{(l)}$ are the learnable weight matrix under relation r and self-loop weight matrix in the l -th layer; $c_{i,r}$ is a normalization constant.

For the link prediction task, we employed a tensor factorization model exploiting the hidden features learned by RGCN layers to calculate the scores of triples. The score of a triple (h, r, t) is calculated as:

$$s(h, r, t) = \mathbf{h}^T \mathbf{r} \mathbf{t}$$

where $\mathbf{h}$, $\mathbf{t}$, $\mathbf{r}$ are the embeddings of h , t , r in Euclidean space. We regarded triples that exist in BKG as positive triples, and generated negative triples by random corrupting of positive triples, which did not exist in BKG before. To ensure that the model assigns higher scores to positive triples than to negative triples, the loss function is defined as [45]:

$$\mathcal{L} = -\frac{1}{(1 + \omega)|\hat{\mathcal{E}}|} \sum_{(h,r,t,y) \in \mathcal{T}} y \log(l(s(h, r, t))) + (1 - y) \log(1 - l(s(h, r, t)))$$

where $\mathcal{T}$ is the triples set (including positive and negative triples); ω is the number of negative triples; $|\hat{\mathcal{E}}|$ is the number of edges; $l(\cdot)$ is the logistic sigmoid function; and y is an indicator, where $y = 1$ means triple is positive, otherwise negative.

2.6. Link prediction evaluation

To evaluate the link prediction performance, we first utilized ranking-based metrics: mean rank (MR), mean reciprocal rank (MRR), and $Hits@k$ ($k = 1, 3$, and 10). For each true triple in the testing set, we generated a set of negative samples by randomly corrupting the head or tail nodes while ensuring that these negative samples do not exist in our graph. We then used the trained model to calculate the scores for the true triple and its negative samples, and obtained the ranks of the true triples. MR , MRR and $Hits@k$ can be calculated based on the ranking of positive triples within their corresponding sets of negative triples. MR represents the average rank assigned to the true triples in the test set:

$$MR = \frac{1}{|T|} \sum_{t \in T} rank(t)$$

where T is all triples in the test set, and $rank(t)$ is the position of the triple t in the sorted list of t and its negative samples.

MRR is the average inverse rank of all true triples in the test set:

$$MRR = \frac{1}{|T|} \sum_{t \in T} \frac{1}{rank(t)}$$

$Hits@k$ is the percentage of triples in which the true triple appears in the top k ranked triples:

$$Hits@k = \frac{1}{|T|} \sum_{t \in T} I[rank(t) \leq k]$$

where I is an indicator function. $I[rank(t) \leq k]$ is equal to 1 if t is ranked between 1 and k , 0 otherwise.

In addition to ranking-based metrics, we used Receiver Operating Characteristic (ROC) curves and Area Under the Receiver Operating Characteristic Curve (AUROC) to evaluate each model on four BKGs. We generated a corresponding negative sample, i.e., the triples that do not exist in the entire BKG, for each triple in the test set by randomly corrupting, thus obtaining a test set with a 1:1 ratio of positive and negative samples [46]. That is, in our study, the triples existing in the test set are labeled as positive, and the generated negative triples are labeled as negative.

3. Results

3.1. Experimental setup

The hyperparameters for TransE, DistMult, and ComplEx were tuned using a grid search on the validation set for each prediction model. We adjusted the following parameters: learning rate $\{0.01, 0.001\}$, number of hidden dimensions $\{100, 200\}$, and regularization coefficient $\{1 \times 10^{-6}, 1 \times 10^{-9}\}$. The mini-batch size was set to $\{250, 1000\}$. In the cases of vanilla RGCN and RGCN layers within FuseLinker's GNN model module, we conducted tuning on the learning rate $\{0.01, 0.001\}$, number of hidden dimensions $\{100, 200\}$, number of GNN layers $\{1, 2\}$, and maintained a mini-batch size of $\{250, 500, 1000\}$. We applied a dropout layer with a rate of 0.2 to the GNN encoder to prevent overfitting and introduced l2 regularization to the link prediction decoder with a penalty of 0.01. For FuseLinker, we experimented with setting weights to $\{0.0, 0.25, 0.50, 0.75, 1.0\}$ for embedding fusion. For all models, negative samples were generated by randomly corrupting the heads or tails of positive triples at a 1:20 ratio during the training process.

3.2. Graph embedding and link prediction evaluation

Table 1 presents the results of our experiments comparing our methods with three traditional knowledge graph embedding methods (TransE, DistMult, and ComplEx), as well as an RGCN link prediction model, across four datasets (KEGG50k, Hetionet, SuppKG, and ADInt). Ranking-based metrics are employed, where a well-performing model should demonstrate a low MR (Mean Rank) score and high MRR (Mean Reciprocal Rank) and Hit@k scores. The p-values from the two-sample t-tests, which compare the performance of FuseLinker with the baseline models, are presented in Table S2 of the supplementary materials. From Table 1, we can observe that across all four datasets, our proposed FuseLinker model consistently outperforms the other models on all ranking-based metrics. For the ADInt dataset, the performance of FuseLinker shows the most significant improvement compared to RGCN. This indicates that our method can outperform vanilla RGCN when dealing with a BKG characterized by a larger number of nodes, edges, and a more intricate topology.

Table 1

Comparing Fuselinker with baseline link prediction methods; Metrics with (↑) indicate higher values are better, while those with (↓) suggest lower values are preferable.

		KEGG50k	Hetionet	SuppKG	ADInt
TransE	MR (↓):	3.240	8.140	6.936	12.165
	MRR (↑):	0.848	0.484	0.641	0.266
	Hits@1 (↑):	0.789	0.339	0.516	0.153
	Hits@3 (↑):	0.893	0.550	0.726	0.271
	Hits@10 (↑):	0.952	0.789	0.873	0.480
DistMult	MR (↓):	7.996	9.029	16.781	13.367
	MRR (↑):	0.761	0.449	0.587	0.215
	Hits@1 (↑):	0.692	0.302	0.523	0.105
	Hits@3 (↑):	0.805	0.513	0.611	0.195
	Hits@10 (↑):	0.886	0.760	0.707	0.473
ComplEx	MR (↓):	6.999	9.272	16.870	12.916
	MRR (↑):	0.770	0.450	0.587	0.225
	Hits@1 (↑):	0.699	0.304	0.525	0.110
	Hits@3 (↑):	0.817	0.513	0.610	0.212
	Hits@10 (↑):	0.899	0.760	0.705	0.486
RGCN	MR (↓):	1.609	7.137	4.275	5.418
	MRR (↑):	0.926	0.533	0.678	0.518
	Hits@1 (↑):	0.891	0.404	0.566	0.383
	Hits@3 (↑):	0.954	0.593	0.745	0.568
	Hits@10 (↑):	0.985	0.797	0.899	0.828
FuseLinker	MR (↓):	1.126	6.424	2.781	1.825
	MRR (↑):	0.969	0.548	0.739	0.831
	Hits@1 (↑):	0.950	0.413	0.627	0.744
	Hits@3 (↑):	0.989	0.615	0.818	0.903
	Hits@10 (↑):	0.998	0.823	0.952	0.983

Table 2

Performance comparison of FuseLinker using text embeddings from various LLMs; Metrics with (↑) indicate higher values are better, while those with (↓) suggest lower values are preferable.

		KEGG50k	Hetionet	SuppKG	ADInt
FuseLinker with BERT embeddings	MR (↓):	1.174	6.640	2.927	2.040
	MRR (↑):	0.962	0.539	0.727	0.785
	Hits@1 (↑):	0.938	0.405	0.611	0.673
	Hits@3 (↑):	0.985	0.604	0.808	0.876
	Hits@10 (↑):	0.997	0.814	0.948	0.981
FuseLinker with PubMedBERT embeddings	MR (↓):	1.126	6.424	2.856	1.825
	MRR (↑):	0.969	0.548	0.728	0.831
	Hits@1 (↑):	0.950	0.413	0.610	0.744
	Hits@3 (↑):	0.989	0.615	0.809	0.903
	Hits@10 (↑):	0.998	0.823	0.951	0.983
FuseLinker with Flan-T5 embeddings	MR (↓):	1.151	6.567	2.781	2.023
	MRR (↑):	0.967	0.542	0.739	0.794
	Hits@1 (↑):	0.945	0.407	0.627	0.688
	Hits@3 (↑):	0.989	0.610	0.818	0.880
	Hits@10 (↑):	0.997	0.816	0.952	0.980
FuseLinker with Llama2 embeddings	MR (↓):	1.147	6.618	2.883	1.984
	MRR (↑):	0.965	0.542	0.732	0.808
	Hits@1 (↑):	0.942	0.408	0.618	0.711
	Hits@3 (↑):	0.987	0.607	0.810	0.887
	Hits@10 (↑):	0.997	0.817	0.948	0.978
FuseLinker with PMC-LLaMA embeddings	MR (↓):	1.146	6.649	2.939	2.066
	MRR (↑):	0.965	0.541	0.718	0.788
	Hits@1 (↑):	0.943	0.408	0.598	0.679
	Hits@3 (↑):	0.987	0.606	0.801	0.875
	Hits@10 (↑):	0.998	0.811	0.947	0.978

Table 2 compares the experimental results of FuseLinker using pre-trained text embeddings from five different LLMs in four datasets. For KEGG50k, SuppKG, and ADInt, FuseLinker achieves the best performance using PubMedBERT text embeddings. However, for SuppKG, FuseLinker achieves the best performance using Flan-T5 text embeddings (MR=2.781, Hits@10 = 0.952), followed by PubMedBERT (MR=2.856, Hits@10 = 0.951) and Llama2 text embeddings (MR=2.883, Hits@10 = 0.948). Table S3 in the supplementary materials presents the p-values from two-sample t-tests comparing the performance of FuseLinker leveraging optimal text embeddings with using text embeddings from other LLMs.

Fig. 2 displays the Receiver Operating Characteristic (ROC) curves of the five models across the four datasets, providing a further comparison of their performance. On the KEGG50k, Hetionet, SuppKG, and ADInt datasets, FuseLinker achieved the highest Area Under the ROC Curve (AUROC), with scores of 0.987, 0.903, 0.928, and 0.890, respectively. Consistent with the results of ranking-based metrics, the AUROC also demonstrates that FuseLinker outperforms the other models.

3.3. t-SNE visualization for BKG embeddings

Subsequently, we utilized t-SNE (t-distributed stochastic neighbor embedding) to obtain the two-dimensional projection of the learned node representations to visualize and compare the quality of the BKG embeddings generated by vanilla RGCN and FuseLinker. Fig. 3 shows the t-SNE visualization results of four BKGs, illustrating the similarity between concepts through the proximity of nodes. For KEGG50k, SuppKG, and ADInt, the clustering distinctions of FuseLinker based on the node types appear more pronounced compared to vanilla RGCN, with particularly notable results on ADInt. Moreover, FuseLinker forms more clusters regardless of the node type, suggesting that the model may have uncovered novel patterns in the data. For Hetionet, the t-SNE visualization results of the vanilla RGCN embeddings are satisfactory. However, it can be observed that some sporadic Compound nodes incorrectly fall into the cluster of Gene nodes. Notably, this issue is absent in the visualization results of FuseLinker. The results of the t-SNE visualizations for all four BKGs indicate that the quality of the graph embeddings obtained by FuseLinker is superior to that of vanilla RGCN.

3.4. Exploring different fusion weights

Fig. 4 illustrates the impact of varying the fusion weight (Hyperparameter w) between pre-trained text embeddings and domain knowledge embeddings on FuseLinker's performance across four BKG datasets. At $w = 0$, the fusion consists entirely of domain knowledge embeddings, whereas at $w = 1$, it comprises solely pre-trained text embeddings. We experimented with setting w to {0.0, 0.25, 0.50, 0.75, 1.0}. Our results indicate that for Hetionet, SuppKG, and ADInt, FuseLinker achieved optimal performance at $w = 0.75$. For KEGG50k, the best performance was observed at $w = 0.25$. These findings confirm that FuseLinker performs more effectively with a balanced integration of both domain knowledge and text information, rather than relying exclusively on one type of embedding.

4. Case studies

To further validate the efficacy of FuseLinker, we conducted case studies on computational drug repurposing to demonstrate the feasibility and practical applications of our approach. Computational drug repurposing aims to identify new therapeutic uses for existing drugs through computer-based tools and data analysis [47]. This method compensates for the shortcomings of traditional drug development, which is often time-consuming and requires substantial manpower and financial resources. One method of computational drug repurposing is to discover novel (Drug, TREATS, Disease) triples by knowledge graph completion or link prediction in BKG to reveal potential other therapeutic effects of known drugs.

In our study, we utilized Hetionet to predict (a) diseases which can be potentially treated by Sorafenib, a targeted cancer drug, and (b) drugs that can potentially treat Parkinson's disease. Using the drug repurposing of Sorafenib as an example, we began by setting the head node to Compound::DB00398 (Sorafenib), assigning relationships as {treats, palliates}, and selecting the tail node from all Disease type nodes in Hetionet, thereby forming a series of triples. Subsequently, we employed the FuseLinker model trained on Hetionet to compute scores for these triplets. After filtering out triplets already present in Hetionet, we sorted the remaining ones and further evaluated the high-scoring triplets. To explore drug repurposing for Parkinson's disease, we set the triple head

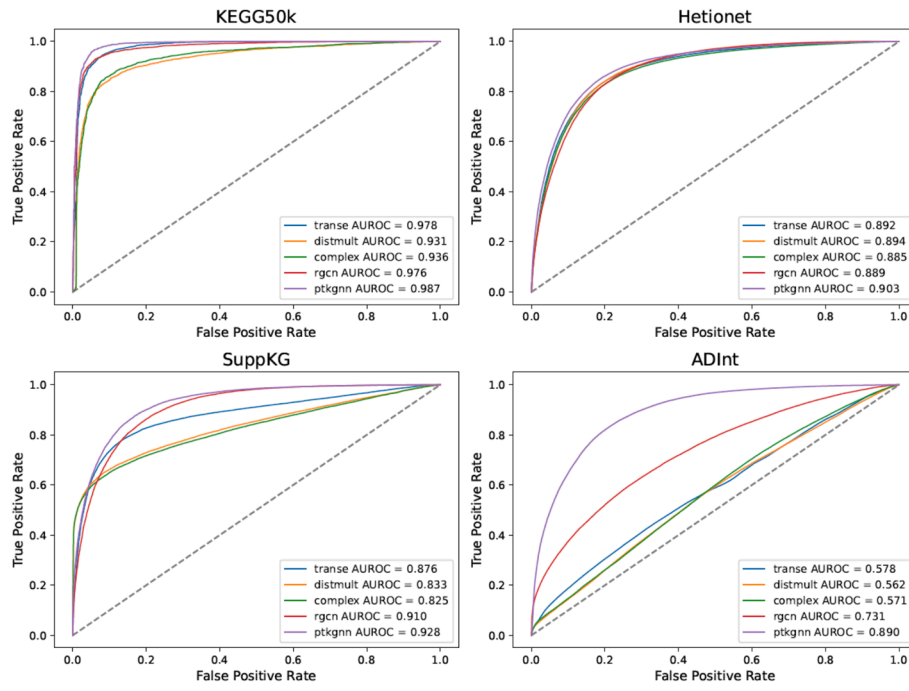


Fig. 2. Receiver Operating Characteristic (ROC) curves and Area Under the ROC Curve (AUROC) of five graph embedding models on four BKGs.

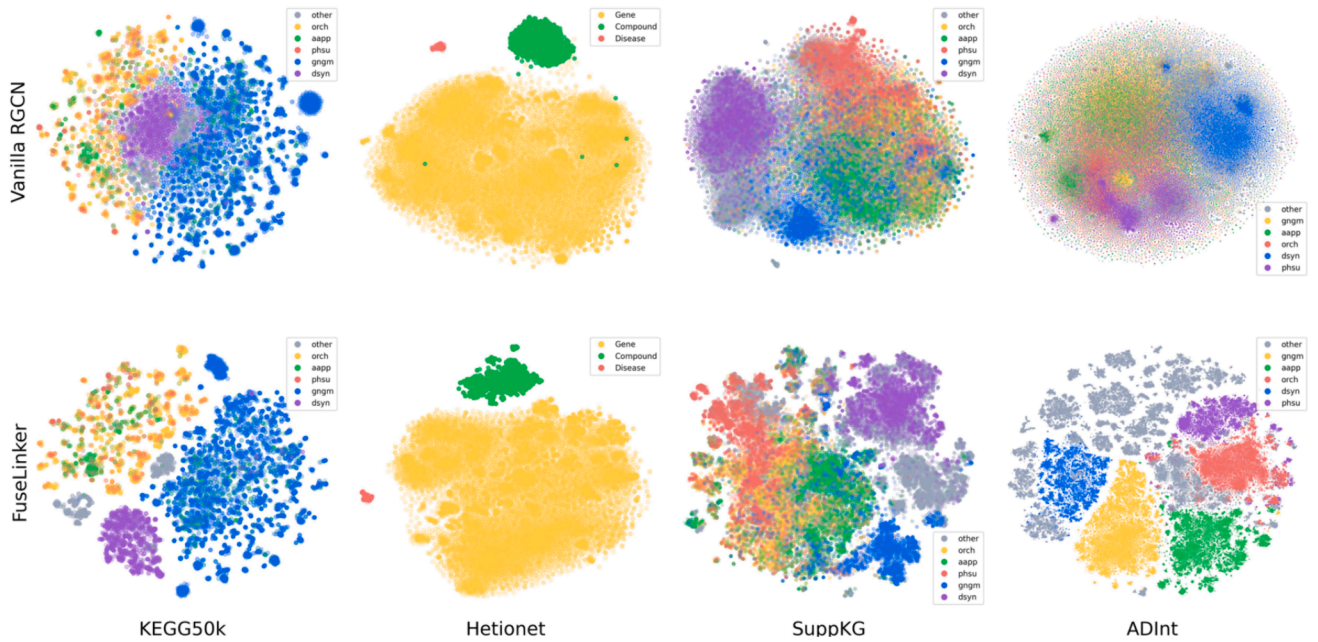


Fig. 3. t-SNE of graph embeddings of Vanilla RGCN and FuseLinker on four BKGs.

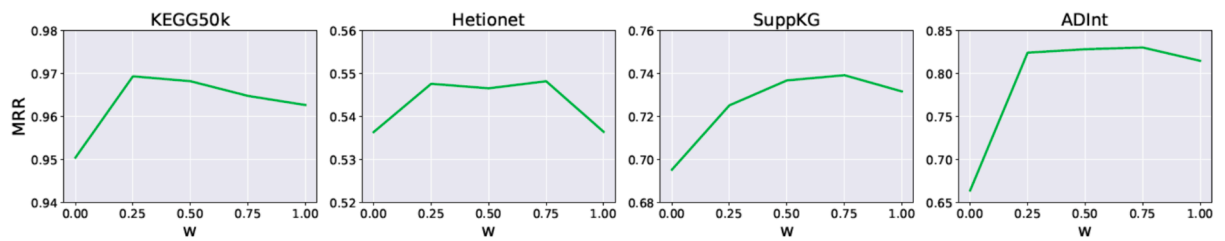


Fig. 4. Impact of fusion weight hyperparameter w on the MRR of FuseLinker across four BKGs.

Table 3

Sorafenib's drug repurposing and drug repurposing for Parkinson's disease on Hetionet.

Predicting diseases treatable with Sorafenib			
	Candidate diseases	Evidence source	PMID or NCT number
1	Lymphatic system cancer	Clinical trial	NCT00474929
2	Breast cancer	Clinical trial	NCT00101400
3	Lung cancer	Clinical trial	NCT00098254
4	Hematologic cancer	Clinical trial	NCT02270788
5	Melanoma	Clinical trial	NCT00483301
6	Kidney cancer	Clinical trial	NCT01982097
7	Multiple sclerosis	Literature evidence	21847523
8	Malaria	Literature evidence	31065106
9	Liver cancer	Potential associations	34166565
10	Coronary artery disease	Potential associations	22687270

Predicting compounds for treating Parkinson's disease.			
	Candidate compounds	Evidence source	PMID or NCT number
1	Sildenafil	Clinical trial	NCT02162979
2	Omeprazole	Clinical trial	NCT00664209
3	Deferoxamine	Literature evidence	32926630
4	Meloxicam	Literature evidence	22617635
5	Tamsulosin	Literature evidence	36054705
6	Artemether	Potential associations	30196102
7	Ranitidine	Potential associations	24379672
8	Cisapride	Potential associations	9159740
9	Fosphenytoin	Potential associations	36578637
10	Delavirdine	No evidence found	

node to encompass all Compound type nodes in Hetionet. The relationship is defined as {treats, palliates}, and the tail node is specified as Disease::DOID:14330 (Parkinson's disease) to construct candidate triples. We utilize the FuseLinker model trained on Hetionet for this task.

Under the open-world assumption, novel triples with high scores that do not exist in the original BKG may represent unknown true facts. To further evaluate the validity of these high-scoring novel triples, we conduct a detailed investigation using various types of evidence, including existing clinical trials and published literature. Specifically, we first checked for completed clinical trials related to the interventions and diseases in the triple using [ClinicalTrials.gov](https://clinicaltrials.gov). If no clinical trials were found, we proceeded to search for published literature on related interventions and diseases in PubMed. If literature was available, we further categorized it into two types [30]: (1) Literature Evidence, which included studies where outcomes and causal relationships between interventions and diseases were explicitly stated; (2) Potential Associations, which included literature that discussed possible, uncertain, or contrary physiological effects, thereby suggesting a less definitive connection between the interventions and disease outcomes. Each high-scoring novel triples was evaluated and qualified by a biostatistics graduate student (SZ) and two health informatics graduate students (YX and HZ) with biomedical, biochemistry, and clinical backgrounds. Table 3 presents the results of two case studies, including the final prediction outcomes, the NCT number of associated clinical trials, and the PMID (PubMed ID) of literature evidence and potential association.

To gain insight into how our model generated novel link predictions, we selected the top predicted drug-disease pairs and analyzed their possible mechanisms based on Hetionet. Fig. 5 illustrates how Sorafenib works in the treatment of predicted Lung cancer and how predicted Sildenafil works in the treatment of Parkinson's disease on Hetionet.

5. Discussion

We proposed a novel link prediction approach, FuseLinker, which innovatively integrates BKG's text information and domain knowledge information. Further, we investigated the effects of utilizing different embedding-visible LLMs and adjusting fusion weights of embeddings on FuseLinker's performance. Through two drug repurposing case studies, we demonstrated FuseLinker's practical application capabilities in real-world scenarios. Inspired by these findings, FuseLinker is also promising for other biomedical applications that fit the link prediction model framework, such as adverse drug reactions and drug-drug interaction detection. These applications could be explored in future work.

Introducing LLM pre-trained text embeddings and domain knowledge pre-trained embeddings significantly enhances the performance of GNN-based link prediction models across all rank-based metrics (MR, MRR and Hits@k). Especially in more complex BKGs such as ADInt, FuseLinker significantly outperforms Vanilla RGCN. This improvement is clearly evidenced in the t-SNE graph of the ADInt embeddings, where FuseLinker demonstrates markedly superior performance. According to Fig. 2, FuseLinker achieved the best ROC and the highest AUROC among the five models across all four datasets, further underscoring its exceptional performance.

We observed that FuseLinker achieves optimal performance with PubMedBERT embeddings on the KEGG50k, Hetionet, and ADInt datasets, and with Flan-T5 embeddings on the SuppKG dataset. We were surprised to find that FuseLinker did not perform very well when using Llama2 and PMC-LLaMA embeddings, despite these models demonstrating good performance in many generative clinical NLP tasks. This underperformance may be attributed to several factors. Firstly, the decoder-only model, such as Llama2 and PMC-LLaMA, may not be optimal for generating text embeddings alone due to the absence of an encoding process for tokens. Additionally, while Llama2 and PMC-LLaMA rely on multi-parameter and high-dimensional embeddings for good results in many NLP tasks, FuseLinker significantly reduces the embedding dimension from 4096 to 200 for these models. In contrast, the reduction is less severe for BERT, PubMedBERT, and Flan-T5, from 768 to 200 dimensions. This greater loss of information might diminish the advantages offered by the larger LLMs. FuseLinker performs better across the board when using PubMedBERT text embeddings compared to BERT text embeddings. This improvement is attributed to PubMedBERT being trained on the PubMed corpus, which enables it to better understand and represent the biomedical and clinical entities, thus providing higher quality text embeddings.

We found that FuseLinker does not achieve the best performance when text embedding and domain knowledge embedding are used

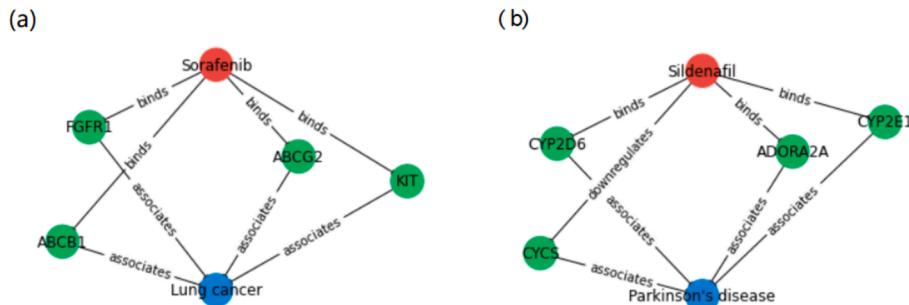


Fig. 5. The possible mechanistic pathways of (a) Sorafenib in the treatment of Lung cancer and (b) Sildenafil in the treatment of Parkinson's disease on Hetionet.

individually (i.e., when w equals 0 or 1). This indicates the significance of integrating text embedding and domain knowledge embedding, as they complement each other and yield an optimal embedding through weighted combination. We compared the results by setting w to 0.0, 0.25, 0.5, 0.75, and 1.0. However, it's possible that other values of w could lead to better performance for FuseLinker. Due to constraints in computing resources and time, further exploration of alternative w values will be pursued in future work.

In the drug repurposing case studies, we predicted the diseases that Sorafenib can treat, which correspond to predicting the tail node through the head node and relationship. Likewise, predicting drugs that can treat Parkinson's disease corresponds to predicting the head node through the node and relationship in the link prediction task. Due to constraints in manpower and time, we manually evaluated the top 10 candidates in each case study. For Sorafenib, among the top 10 predicted candidate diseases, 6 diseases (60 %) have completed clinical trials with Sorafenib, 2 diseases (20 %) have been verified by literature evidence to be treatable by Sorafenib, and 2 candidate diseases (20 %) have a potential relationship with Sorafenib. Similarly, for Parkinson's disease, among the top 10 predicted candidate drugs, 2 drugs (20 %) have completed clinical trials with Parkinson's disease, 3 drugs (30 %) have been verified by literature evidence to treat Parkinson's disease, and 4 candidate drugs (40 %) have potential associations with Parkinson's disease. Notably, we found no clinical trials or literature evidence supporting the use of Delavirdine to treat Parkinson's disease. Additionally, leveraging Hetionet, we presented the mechanistic pathways of Sorafenib in the treatment of Lung cancer and Sildenafil in the treatment of Parkinson's disease. These mechanistic pathways are entirely derived from Hetionet, but they are further supported by relevant evidence from published literature. For instance, Sorafenib has been observed to inhibit FGFR1 gene activity in mesothelioma tumor-initiating cells, which suggests a potential role in blocking the autocrine signaling loops involving this receptor [48]. FGFR1 is frequently involved in lung cancer, particularly in squamous cell carcinoma. It can act as a predictive biomarker for the response to FGFR tyrosine kinase inhibitors, which are used in treating lung cancer [49]. Therefore, these pieces of evidence support the potential of sorafenib to treat lung cancer via FGFR1 inhibition from an in silico drug repurposing perspective.

Our study also has some limitations and can be improved in future work. Firstly, while FuseLinker utilizes well-known LLMs like BERT, PubMedBERT, Flan-T5, Llama2 and PMC-LLaMA models to obtain pre-trained text embeddings, their other variants like BlueBERT, med-Alpaca, and newer LLMs like Mamba and Mistral also have the potential to produce high-quality text embeddings and can be further studied in the future. Secondly, our use of MetaMap to map entities in BKGs to the UMLS Semantics network has limitations, as studies have shown MetaMap's performance is not flawless [50]. Future work could explore more accurate methods for mapping biomedical entity names to the UMLS Metathesaurus. Additionally, the multi-relation graph neural network module of FuseLinker currently employs traditional RGCN layers for processing heterogeneous graph structure information. Investigating and experimenting with alternative graph convolutional layers may enhance FuseLinker's performance. Lastly, in our two case studies, due to constraints in human resources and time, we only evaluated the top ten hypotheses with the highest scores. Future efforts could extend to evaluating more prediction candidates and conducting further real-world data analyses.

6. Conclusion

In this work, we introduced FuseLinker, a novel GNN-based link prediction method with multi-information fusion. FuseLinker demonstrates notable improvements over traditional KGE models and vanilla GNN models in the link prediction task across four BKG datasets. This enhancement is driven by the introduction and fusion of LLM's text embeddings and pre-trained domain knowledge embeddings. To the

best of our knowledge, this is the first study to combine graph structural, textual, and domain knowledge information for link prediction on BKGs. In two drug repurposing case studies, we showcased the practical applicability of FuseLinker, where we manually evaluated high-scoring hypotheses and explored generative mechanistic explanations. Looking ahead, our approach holds significant promise for addressing other graph-based biomedical link prediction tasks, thereby underscoring the broad applicability and importance of our research in the biomedical field.

Fundings

This work was supported by the National Institutes of Health's National Center for Complementary and Integrative Health under grant number R01AT009457, National Institute on Aging under grant number R01AG078154 and National Cancer Institute under grant number R01CA287413. The content is solely the responsibility of the authors and does not represent the official views of the National Institutes of Health.

CRediT authorship contribution statement

Yongkang Xiao: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sinian Zhang:** Writing – original draft, Validation, Software, Investigation, Data curation. **Huixue Zhou:** Writing – review & editing, Validation. **Mingchen Li:** Writing – review & editing. **Han Yang:** Writing – review & editing. **Rui Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We extend our sincere gratitude to the Minnesota Supercomputing Institute at the University of Minnesota for the computing resources provided, which were invaluable to our research.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jbi.2024.104730>.

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